Hardware-Aware $SE(3)$ Control Barrier Functions for Counter-UAS Interceptors with Directed Energy Payloads

Evangelos Vlachos[1], Panayiotis Kolios[2], and Christos Skliros[3]

***Abstract*— Strict power limits on small drone interceptors constrain the use of reusable directed energy payloads such as RF jammers and High Power Microwave (HPM) sources. High-gain beams can close the engagement link budget with minimal transmit power, but their required tight Field of View (FoV) conflicts with the aggressive maneuvers of a pursuing multi-rotor. This paper resolves this conflict with a hardware-aware safety filter using Control Barrier Functions (CBFs) on the $SE(3)$ manifold. The filter provides formal pointing guarantees, enabling highly directional antennas whose gain would otherwise be sacrificed for pointing tolerance. We formulate a phased array's electronic steering limits as a relative-degree-two safety constraint with analytically verified Lie derivatives and solve the resulting Quadratic Program (QP) via ADMM with a constant-time core factorization. Validation in a nonlinear $SE(3)$ simulation and a custom 6-DOF Software-in-the-Loop (SITL) environment demonstrates the proposed CBF eliminates all FoV violations and recovers up to 12.0 dB of antenna gain by tightening the beam. The approach also reveals conventional pitch clamping provides negligible benefit against FoV violations, avoids the severe actuator-saturation failure modes of multi-step Model Predictive Control (MPC)-CBF architectures, and executes reliably in under 350 μs.**

Index Terms—Counter-UAS, Control Barrier Functions, $SE(3)$, directed energy, field of view, ADMM, quadrotor, safety filter.

I. Introduction

The proliferation of rogue Unmanned Aerial Vehicles (UAVs) has accelerated the deployment of Counter-UAS (C-UAS) interceptors [1]. Directed energy payloads, such as high-power radio frequency (HPRF) and microwave systems, provide reusable non-kinetic defeat mechanisms [2]. However, transitioning these power-hungry, kilowatt-class systems to battery-constrained small interceptors requires a paradigm shift in power management. Payload effectiveness scales directly with antenna gain, which is inversely proportional to the beam's field of view (FoV). Narrowing the steerable FoV from a conservative 60° to 23° recovers 8.3 dB of gain, reducing the required transmit power by a factor of $6.8\times$. Yet, a severe pointing conflict emerges: aggressive pursuit maneuvers induce large coupled attitude transients in underactuated multi-rotor kinematics [3], inevitably rotating rigidly attached payloads away from the target.

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Background on CBFs. A Control Barrier Function (CBF) is a scalar function $h(\mathbf{x})$ whose superlevel set $\mathcal{S} = \{h \geq 0\}$ defines a safe set. A CBF safety filter modifies a nominal controller minimally at each step to keep the system inside $\mathcal{S}$ [8]. When h has relative degree greater than one with respect to the control input, as is the case for attitude-coupled pointing constraints, an Exponential CBF (ECBF) is required: additional Lie derivative conditions extend the CBF constraint to the appropriate derivative order [16]. ECBFs generalize High Order CBFs (HOCBFs) [9] in that both handle relative-degree-r constraints, but ECBFs use an exponential decay class for the auxiliary conditions, yielding a closed-form linear constraint that is particularly efficient to solve.

Existing UAV tracking controllers frequently treat interceptors as kinematic point-masses [4], focus on vision-based collision avoidance [5], [6], or rely on mechanical gimbals [7]. Without formal geometric pointing guarantees, designers must widen the FoV to accommodate uncertainty, sacrificing crucial antenna gain. While visual servoing CBFs exist [10], they operate entirely in approximated image-plane coordinates. In contrast, our formulation operates directly on the $SE(3)$ attitude manifold using analytically derived Lie derivatives of a body-frame pointing constraint, precisely matching phased array physics where exceeding the FoV envelope causes immediate RF signal loss [18].

Finally, ensuring deterministic execution for high-frequency flight loops remains a critical challenge. While recent literature has successfully combined Model Predictive Control with Control Barrier Functions (MPC-CBF) to optimize safety over a future horizon [11], [12], these multi-step predictive filters introduce severe computational bottlenecks and actuator saturation risks when applied to underactuated 3D kinematics. Furthermore, standard interior-point QP solvers suffer from variable iteration counts that destabilize real-time control. We overcome this by deploying an Alternating Direction Method of Multipliers (ADMM) solver [13]–[15]. Our slack-relaxed ADMM safety filter leverages the fixed low dimensionality of the actuation space to guarantee $\mathcal{O}(1)$ constant-time Cholesky factorization at each filter step, avoiding the actuator saturation and computational bottlenecks that afflict MPC-CBF architectures.

A. Statement of Contributions

To enable drone-mounted directed energy by providing the rigid pointing guarantees that unlock tight, power-efficient beam designs, the core novelties of this paper are:

- *Hardware-aware $SE(3)$ barrier formulation*: We map the rigid electronic steering limits of phased array payloads directly onto the $SE(3)$ manifold as a relative-degree-two ECBF. Unlike traditional image-plane visual servoing, this provides an analytically exact Lie derivative chain that directly governs the physical RF link budget.
- *Deterministic $\mathcal{O}(1)$ ADMM safety filter*: We propose a slack-relaxed ADMM solver for the CBF-QP whose Cholesky factorization operates on a fixed-size system, guaranteeing constant-time matrix operations at each step. This enables microsecond-scale execution suitable for high-rate inner-loop flight controllers, with predictable per-iteration cost and effective warm-starting that keeps the mean iteration count low.
- *Antenna gain recovery and predictive benchmarking*: Through rigorous PX4-compatible SITL and nonlinear simulations, we quantify up to 12.0 dB of gain recovery (a 16× transmit power reduction) via beam tightening. Furthermore, we expose a fundamental actuator-saturation failure mode in standard MPC-CBF architectures and demonstrate empirically that conventional pitch-clamping heuristics provide negligible reduction in FoV violations due to coupled rotational dynamics.

II. Problem Formulation

A. Quadrotor Kinematics and Dynamics on $SE(3)$

Consider a quadrotor operating in a 3D airspace. Let $\mathbb{R}^3$ denote the Euclidean space and $SO(3)$ the Special Orthogonal group. The state is defined by position $\mathbf{p} \in \mathbb{R}^3$, velocity $\mathbf{v} \in \mathbb{R}^3$, attitude rotation matrix $\mathbf{R} \in SO(3)$ (mapping the body frame $\mathcal{B}$ to the inertial frame $\mathcal{W}$), and body angular velocity $\boldsymbol{\Omega} \in \mathbb{R}^3$. The nominal control $\mathbf{u}_{\text{nom}} = [f_{\text{nom}}, \boldsymbol{\tau}_{\text{nom}}^T]^T$ is provided by the Lee geometric trajectory tracker [17], which generates thrust and torque commands to follow a reference trajectory. The CBF safety filter modifies these commands minimally at each step to enforce the pointing constraint.

The rigid-body equations of motion are given by:

$$\dot{\mathbf{p}} = \mathbf{v}, \tag{1}$$

$$m\dot{\mathbf{v}} = -mg\mathbf{e}_3 + f\mathbf{R}\mathbf{e}_3, \tag{2}$$

$$\dot{\mathbf{R}} = \mathbf{R}\hat{\boldsymbol{\Omega}}, \tag{3}$$

$$\mathbf{J}\dot{\boldsymbol{\Omega}} = \boldsymbol{\tau} - \boldsymbol{\Omega} \times \mathbf{J}\boldsymbol{\Omega}, \tag{4}$$

where m is the mass, $\mathbf{J} \in \mathbb{R}^{3\times3}$ is the inertia matrix, g is the gravitational acceleration, $\mathbf{e}_3 = [0, 0, 1]^T$, and $\hat{\cdot}$ denotes the skew-symmetric map. The control inputs are total thrust $f \in \mathbb{R}$ and body torques $\boldsymbol{\tau} \in \mathbb{R}^3$. We define the control vector as $\mathbf{u} = [f, \boldsymbol{\tau}^T]^T \in \mathbb{R}^4$.

B. Directed Energy Payload and FoV Constraint

The directed energy payload (phased array jammer or HPM source) is rigidly mounted on the vehicle with boresight direction $\mathbf{v}_B \in \mathbb{S}^2$ expressed in the body frame $\mathcal{B}$. The phased array can steer its beam within a half-angle cone of $\Phi_{\max}$. A target at position $\mathbf{p}_T \in \mathbb{R}^3$ is within the steerable FoV if and only if the angle between the boresight (expressed in world frame) and the line-of-sight (LOS) unit vector $\bar{\mathbf{n}}$ does not exceed $\Phi_{\max}$:

$$\cos\left(\angle(\mathbf{R}\mathbf{v}_B, \bar{\mathbf{n}})\right) \;\geq\; \cos\Phi_{\max}, \tag{5}$$

where $\bar{\mathbf{n}} = (\mathbf{p}_T - \mathbf{p})/\|\mathbf{p}_T - \mathbf{p}\|$. Equivalently, since $\mathbf{R}$ is orthogonal, this can be expressed in the body frame as:

$$h(\mathbf{R}, \mathbf{p}) \;\triangleq\; \mathbf{v}_B^T\mathbf{R}^T\bar{\mathbf{n}} \;-\; \cos\Phi_{\max} \;\geq\; 0. \tag{6}$$

The set $\mathcal{S} = \{(\mathbf{R}, \mathbf{p}) : h \geq 0\}$ is the *safe set* to be rendered forward-invariant by the CBF filter.

The antenna gain, relative to a reference half-angle Φ_{ref}, follows the solid-angle scaling $G \propto 1/\Phi^2$:

$$\Delta G(\Phi_{\max}) = 20\log_{10}\left(\frac{\Phi_{\text{ref}}}{\Phi_{\max}}\right) \quad [\text{dB}]. \tag{7}$$

Throughout this paper we adopt $\Phi_{\text{ref}} = 60^\circ$ as a conservative baseline, consistent with typical wide-FoV phased array designs. This quantifies the power efficiency benefit of beam tightening.

III. CBF Safety Filter on $SE(3)$

A. Relative Degree Analysis

To apply an ECBF, we compute the relative degree of h with respect to the control inputs $\mathbf{u}$. Differentiating (6):

$$\dot{h} = \mathbf{v}_B^T\left(-\hat{\boldsymbol{\Omega}}\mathbf{R}^T\bar{\mathbf{n}} + \mathbf{R}^T\dot{\bar{\mathbf{n}}}\right). \tag{8}$$

Substituting $\dot{\bar{\mathbf{n}}} = \frac{1}{d}\mathbf{P}_W\dot{\mathbf{p}}$ where $d = \|\mathbf{p}_T - \mathbf{p}\|$ and $\mathbf{P}_W = \mathbf{I} - \bar{\mathbf{n}}\bar{\mathbf{n}}^T$ is the projection onto the plane perpendicular to $\bar{\mathbf{n}}$, we find that $\dot{h}$ depends on $\boldsymbol{\Omega}$ (which is driven by $\boldsymbol{\tau}$) and on $\mathbf{v}$ (which is driven by f). However, $\dot{h}$ contains $\boldsymbol{\Omega}$ but not $\boldsymbol{\tau}$ directly. Differentiating again yields:

$$\ddot{h} = L_f^2 h + L_{g_f}L_f h \cdot f + L_{\mathbf{g}_{\boldsymbol{\tau}}}L_f h \cdot \boldsymbol{\tau}, \tag{9}$$

where $L_f^2 h$ is the drift term (Lie derivative along f-vector field twice), $L_{g_f}L_f h$ is the thrust Lie derivative, and $L_{\mathbf{g}_{\boldsymbol{\tau}}}L_f h \in \mathbb{R}^{1\times3}$ is the torque Lie derivative row vector. The linear appearance of the control input $\mathbf{u}$ in $\ddot{h}$ confirms relative degree 2. This relative-degree-two structure is why a standard (relative-degree-one) CBF cannot be applied directly; the ECBF formulation [16] is required to handle the intermediate Lie derivative $\dot{h}$.

B. Lie Derivative Expressions and ECBF Constraint

Let $\mathbf{n}_b = \mathbf{R}^T\bar{\mathbf{n}}$ be the LOS direction in body frame. The analytical Lie derivatives are:

$$L_{\mathbf{g}_{\boldsymbol{\tau}}}L_f h = (\mathbf{v}_B \times \mathbf{n}_b)^T\mathbf{J}^{-1}, \tag{10}$$

$$L_{g_f}L_f h = -\frac{1}{md}\,\mathbf{v}_B^T\mathbf{R}^T\mathbf{P}_W\mathbf{R}\mathbf{e}_3, \tag{11}$$

$$\begin{aligned} L_f^2 h = {} & (\mathbf{v}_B \times \mathbf{n}_b)^T\mathbf{J}^{-1}(-\boldsymbol{\Omega} \times \mathbf{J}\boldsymbol{\Omega}) \\ & + (\mathbf{v}_B \times \dot{\mathbf{n}}_b)^T\boldsymbol{\Omega} + \mathbf{v}_B^T\mathbf{R}^T\ddot{\bar{\mathbf{n}}}_{\text{drift}}, \end{aligned} \tag{12}$$

where $\ddot{\bar{\mathbf{n}}}_{\text{drift}}$ is the kinematic acceleration of the LOS unit vector due to gravity and the current velocity (no control contribution). These expressions are computed analytically and evaluated in $\mathcal{O}(1)$ at each filter step.

The ECBF condition for relative-degree-2 systems requires:

$$L_f^2 h + L_{g_f} L_f h \cdot f + L_{\mathbf{g}_\tau} L_f h \cdot \boldsymbol{\tau} \;\geq\; -c_1 \dot{h} - c_2 h - \delta, \quad (13)$$

where $c_1, c_2 > 0$ are chosen so that the characteristic polynomial $s^2 + c_1 s + c_2$ has roots in the open left half-plane, ensuring forward invariance [16]. The gains $c_1 = 8$ and $c_2 = 16$ yield a critically-damped response (roots at $s = -4$), maximizing convergence rate without overshoot and validated to be insensitive to $\Phi_{\max}$ (Section V-C). A slack variable $\delta \geq 0$ prevents filter failure when the QP would otherwise be infeasible (e.g., at takeoff), and the large penalty on δ in the objective causes the optimizer to drive $\delta \approx 0$ whenever a feasible safe control exists, preserving the hard safety guarantee in normal operation.

C. Safety Filter QP

The safety filter modifies the nominal control $\mathbf{u}_{\text{nom}}$ minimally subject to the ECBF constraint and actuator limits. The augmented control $\tilde{\mathbf{u}} = [f, \tau_x, \tau_y, \tau_z, \delta]^T \in \mathbb{R}^5$ is found by solving:

$$\begin{aligned} \min_{\tilde{\mathbf{u}}} \quad & \frac{1}{2} \tilde{\mathbf{u}}^T \mathbf{H} \tilde{\mathbf{u}} + \mathbf{q}^T \tilde{\mathbf{u}} \\ \text{s.t.} \quad & \mathbf{A} \tilde{\mathbf{u}} \leq \mathbf{b}, \end{aligned} \quad (14)$$

where $\mathbf{H} = \text{diag}(\mathbf{W}, p_1)$ with $\mathbf{W} = \text{diag}(w_f, w_\tau, w_\tau, w_\tau)$ (w_f and w_τ are the thrust and torque tracking weights, respectively), $\mathbf{q} = [(-\mathbf{W}\mathbf{u}_{\text{nom}})^T, 0]^T$, and $p_1 \gg 1$ is a large penalty on the slack. The constraint matrix $\mathbf{A} \in \mathbb{R}^{10 \times 5}$ and right-hand side $\mathbf{b} \in \mathbb{R}^{10}$ encode:

$$\begin{cases} -L_{\mathbf{g}_\tau} L_f h \cdot \boldsymbol{\tau} - L_{g_f} L_f h \cdot f - \delta \leq L_f^2 h + c_1 \dot{h} + c_2 h, \\ f_{\min} \leq f \leq f_{\max}, \\ |\tau_i| \leq \tau_{\max}, \quad i \in \{x, y, z\}, \\ \delta \geq 0. \end{cases} \quad (15)$$

IV. ADMM Safety Filter Algorithm

The QP (14) is solved via ADMM [19] applied to the consensus form. Introducing an auxiliary variable $\mathbf{z} = \mathbf{A}\tilde{\mathbf{u}}$, the augmented Lagrangian is:

$$\begin{aligned} \mathcal{L}_\rho = & \frac{1}{2} \tilde{\mathbf{u}}^T \mathbf{H} \tilde{\mathbf{u}} + \mathbf{q}^T \tilde{\mathbf{u}} \\ & + \mathbf{y}^T (\mathbf{A}\tilde{\mathbf{u}} - \mathbf{z}) + \frac{\rho}{2} \|\mathbf{A}\tilde{\mathbf{u}} - \mathbf{z}\|^2. \end{aligned} \quad (16)$$

The $\tilde{\mathbf{u}}$-update requires solving a linear system with coefficient matrix $\mathbf{M} = \mathbf{H} + \rho \mathbf{A}^T \mathbf{A}$, which is factored *once* per filter step via Cholesky:

$$\mathbf{M} \triangleq \mathbf{H} + \rho \mathbf{A}^T \mathbf{A} = \mathbf{L}\mathbf{L}^T. \quad (17)$$

The $\mathbf{z}$-update is a projection onto $(-\infty, \mathbf{b}]$, computed element-wise in $\mathcal{O}(k)$ where $k = 10$ is the number of constraints. The dual variable $\mathbf{y}$ is updated with step size ρ. Algorithm 1 summarizes the complete procedure.

The claim of "constant-time" execution refers specifically to the Cholesky factorization and the two triangular back-solves: these operate on a fixed-size 5×5 matrix and require a constant number of operations regardless of the number of constraints k or the problem data. The ADMM iteration count varies between 1 and $i_{\max} = 50$, with a low mean and P95 $= 26$ in our experiments, so the total solve time is *not* constant. It is the core per-iteration cost that remains fixed. This predictability is the key advantage over interior-point solvers, whose iteration count depends on the problem geometry and can spike unpredictably. The reported 347 μs timing (mean total filter step) is measured entirely within the CBF node process; it excludes the uXRCE-DDS bridge latency between the companion computer and the PX4 flight controller, which adds a further 1–2 ms of transport delay at 250 Hz but does not affect the computational safety guarantee. The nonlinear $SE(3)$ simulation runs at 500 Hz (Table I), while the SITL validation runs the filter at 250 Hz.

Algorithm 1 ADMM CBF Safety Filter

Require: State $(\mathbf{R}, \mathbf{p}, \mathbf{v}, \boldsymbol{\Omega})$, nominal $\mathbf{u}_{\text{nom}}$, target $\mathbf{p}_T$
Ensure: Safe control $\mathbf{u}_{\text{safe}}$

```
Compute h, ḣ, L_f^2 h, L_{g_τ} L_f h, L_{g_f} L_f h via (10)–(12)
Assemble H, q, A, b from (14), (15)
Factor M = H + ρ A^T A via Cholesky (17)   (once per step)
Warm-start z, y from previous iterate (or zero)
for i = 1, ..., i_max do
    ũ ← M^{-1}(−q + ρ A^T (z − y/ρ))   (L^{-T} L^{-1} triangular solves)
    z ← min(A ũ + y/ρ, b)   (projection)
    y ← y + ρ(A ũ − z)   (dual update)
    if ‖A ũ − z‖ < ε then
        converged; break
    end if
end for
if not converged then
    ũ ← reset; return u_nom (fallback)
end if
u_safe ← clip(ũ_{1:4}, bounds)
return u_safe
```

V. Simulation Results

A. Experimental Setup

The Lee geometric controller [17] provides the nominal control $\mathbf{u}_{\text{nom}}$. The vehicle follows a four-waypoint pursuit trajectory toward a stationary target. Yaw is commanded toward the current waypoint, deliberately inducing roll-dominant FoV violations during lateral turns, representing the worst-case scenario for pitch-only heuristics.

Four configurations are compared:

(a) **Unfiltered Baseline**: Relies solely on the standard geometric tracker to pursue the target. It provides the unbounded nominal control effort $\mathbf{u}_{\text{nom}}$, establishing the baseline performance and demonstrating how unconstrained 3D pursuit dynamics naturally violate the payload FoV.

(b) **Pitch Clamp**: Represents a common heuristic approach. It independently saturates the body-Y (pitch axis) torque

to prevent the vehicle's nose from tilting beyond $\Phi_{\max}$, while leaving roll and yaw torques completely unconstrained.

(c) **ADMM-CBF (Proposed)**: The single-step safety filter derived in this work. It leverages the exact Lie derivatives of the $SE(3)$ kinematics to reactively modulate all rotational axes simultaneously, rendering the safe set forward-invariant with a constant $\mathcal{O}(1)$ Cholesky factorization at each step.

(d) **MPC-CBF**: A predictive baseline extending (c) with $N = 10$ look-ahead steps ($\Delta t_{\text{pred}} = 0.01$ s). It generates a predicted state trajectory by rolling out the *unfiltered* nominal controller in open loop for N steps, then imposes $N + 1$ simultaneous ECBF constraints (one per predicted state) on a single current control $\tilde{\mathbf{u}}$ with shared slack δ. This is a single-control filtering problem, not a full MPC sequence, as discussed further in Section V-B.

TABLE I

SE(3) SIMULATION PARAMETERS.

Parameter	Symbol	Value
Mass	m	1.0 kg
Inertia (diag.)	$\mathbf{J}$	$[0.01, 0.01, 0.02]$ kg m^2
Min thrust	$f_{\min}$	0 N
Max thrust	$f_{\max}$	25 N
Max torque	$\tau_{\max}$	2 N m
FoV half-angle	$\Phi_{\max}$	25°
ECBF gains	c_1, c_2	8, 16
ADMM penalty	ρ	10
Slack penalty	p_1	10^5
Cost weights	$\mathbf{W}$	$\text{diag}(1, 10, 10, 10)$
Boresight (body)	$\mathbf{v}_B$	$[1, 0, 0]^T$
Filter rate	–	500 Hz

B. Four-Way Comparison

Table II and Fig. 1 compare all four configurations over a 25 s pursuit scenario. The ADMM-CBF eliminates all FoV violations while maintaining safe boresight pointing ($h_{\min} = -0.001$, within numerical integration tolerance) and tracking all four waypoints. The pitch-clamp heuristic yields a violation rate of 12.4% versus 12.3% for the unfiltered baseline, a negligible difference that confirms pitch-axis saturation is ineffective when violations arise from roll/yaw misalignment during lateral turns.

The oscillations visible in the attitude states and h_{FoV} traces of Fig. 1 arise from the warm-start mechanism: after each waypoint transition, the ADMM dual variables are re-initialized from the previous iterate, which was computed at a substantially different operating point. This cold-restart transiently widens the residual and drives a brief burst of corrective attitude adjustments before convergence is reestablished. The effect is especially pronounced for the MPC-CBF baseline because its $N + 1$ simultaneous constraints amplify the recovery demand at each restart.

The MPC-CBF achieves a strictly positive safety margin ($h_{\min} = +0.084$), but at a significant mission cost. Because the predicted rollout uses the *open-loop, unfiltered* nominal controller rather than the true future filtered trajectory, the predicted states drift pessimistically to $h \approx -0.3$, imposing large simultaneous recovery demands on all $N+1$ constraints at once. A single control $\tilde{\mathbf{u}}$ must satisfy all of them, shrinking the feasible region severely: maximum pitch falls to 5.1° (vs. 18.8° for ADMM-CBF), preventing waypoint tracking within the 25 s window (Fig. 1(a), green). A proper MPC-CBF would require optimizing the full control *sequence* $\mathbf{u}_0, \ldots, \mathbf{u}_{N-1}$, a $4N$-variable nonlinear QP that is computationally intractable at 250 Hz.

The ADMM-CBF sidesteps this entirely: a single reactive ECBF constraint on the current state suffices to render the safe set forward-invariant. The 5×5 factorization is fixed, achieves a 347 μs mean solve time, and never degrades mission tracking.

TABLE II

SE(3) SIMULATION: 4-WAY QUANTITATIVE COMPARISON

($\Phi_{\max} = 25°, T = 25$ s).

Method	min h_{FoV}	Pitch (°)	Roll (°)	Viol. (%)
Unfiltered	-0.345	33.4	12.7	12.3
Pitch clamp	-0.344	33.4	12.7	12.4
ADMM-CBF	-0.001^*	18.8	15.0	**0.0**
MPC-CBF †	$+0.084$	$5.1^\ddagger$	$1.2^\ddagger$	**0.0**

*Within numerical integration tolerance, not a safety violation.
$^\dagger N = 10$, $\Delta t = 0.01$ s. ‡Actuator sat., waypoint tracking degraded.

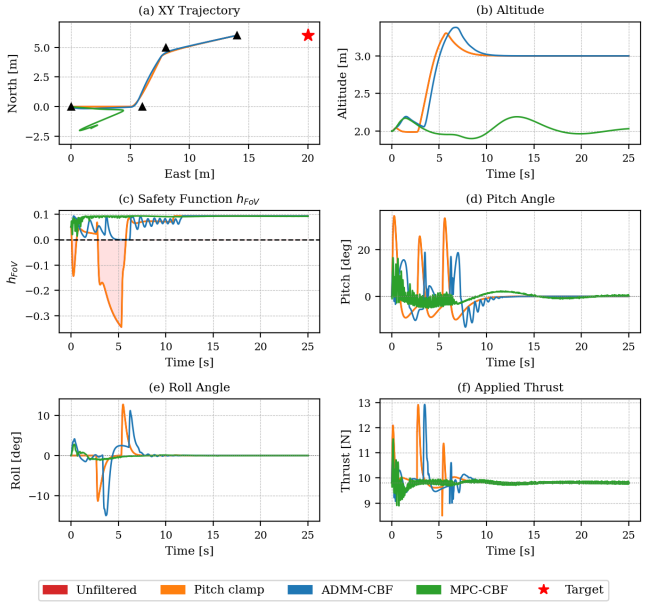


Fig. 1. SE(3) simulation results for the four-way comparison ($\Phi_{\max} = 25°$). All four methods are shown: unfiltered baseline (red), pitch clamp (orange), ADMM-CBF (blue), and MPC-CBF (green). The ADMM-CBF maintains $h_{FoV} \geq 0$ throughout the pursuit and reaches all waypoints. MPC-CBF achieves a larger safety margin ($h_{FoV} \gg 0$) but over-constrains attitude to 5.1° pitch due to open-loop rollout pessimism, preventing waypoint tracking. The pitch clamp and unfiltered baseline show nearly identical violation rates. Shaded regions indicate FoV violation intervals.

C. $\Phi_{\max}$ Sweep and Gain Recovery

Table III and Fig. 2 report the CBF performance as the beam half-angle is tightened from 25° down to 15°. The CBF maintains zero violations across all configurations, confirming robustness across the entire gain-efficiency Pareto frontier. Tightening from 25° to 15° recovers an additional 4.4 dB of antenna gain, demonstrating that the filter safely enables beam tightening to the hardware limit of the phased array. The 15° lower bound is set by the phased array hardware specification, not by the CBF formulation itself. The ECBF constraint imposes no internal minimum on $\Phi_{\max}$: it remains feasible as long as $\mathcal{S}$ is non-empty, i.e., as long as the vehicle can physically orient its boresight toward the target.

TABLE III

$\Phi_{\max}$ SWEEP: CBF PERFORMANCE AND ANTENNA GAIN RECOVERY.

$\Phi_{\max}$ (°)	$\min h_{FoV}$*	Viol. (%)	Gain vs. 60° (dB)
25	−0.0008	0.0	+7.6
23	−0.0007	0.0	+8.3
20	−0.0007	0.0	+9.5
15	−0.0013	0.0	+12.0

*All values within numerical integration tolerance, not violations.

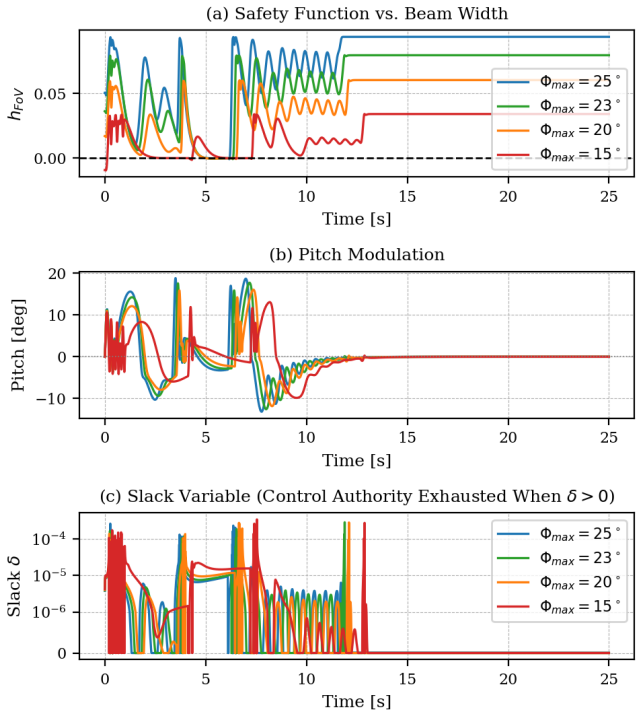


Fig. 2. $\Phi_{\max}$ sweep results. (a) Safety function h_{FoV} remains non-negative for all beam widths. (b) Pitch modulation increases as the constraint tightens, reflecting the filter exerting more corrective authority. (c) Slack variable $\delta > 0$ during brief periods (e.g., takeoff transients) confirms the feasibility relaxation role of δ, which is otherwise inactive ($\delta \approx 0$).

VI. PX4 SITL VALIDATION

A. Platform and Flight-Stack Architecture

The SITL validation uses a closed-loop ROS 2 system composed of three nodes communicating over standard PX4 uORB-bridged topics, with a custom 6-DOF rigid-body physics environment replacing the full PX4+Gazebo simulator for rapid, repeatable offline testing. The three nodes are:

1) **CBF node** (`cbf_node`): runs the ADMM safety filter at 250 Hz. Subscribes to vehicle odometry (pose, velocity, angular rates) and the nominal setpoint from the mission node. Publishes safe body-rate commands to `/fmu/in/vehicle_rates_setpoint` and maintains the offboard heartbeat.
2) **Mission node** (`mission_node`): implements the Lee $SE(3)$ geometric controller [17] at 50 Hz. Subscribes to vehicle odometry, sequences through the four waypoints, and publishes nominal thrust/torque as a `WrenchStamped` to the CBF node. Issues ARM and OFFBOARD vehicle commands once the CBF node heartbeat is established.
3) **Custom Physics Integrator** (`px4_physics`): Euler-integrates 6-DOF rigid-body dynamics at 250 Hz, publishes `VehicleOdometry` and `VehicleStatus` on PX4 uORB topics, and receives `VehicleRatesSetpoint` from the CBF node to close the loop.

All inter-node topics use PX4-compatible QoS (best-effort reliability, volatile durability, depth 1) to faithfully replicate the uXRCE-DDS bridge behavior. The CBF node is launched 5 s before the mission node so that the offboard heartbeat is established before the first ARM command is issued.

The control injection point is the *body-rate* level: the CBF filter outputs $[f, \boldsymbol{\Omega}_{des}]$ rather than attitude setpoints. This bypasses the PX4 attitude controller and applies corrections at the highest bandwidth available on the stack, essential for a 250 Hz safety filter. The desired torque $\boldsymbol{\tau}^*$ is converted to a rate setpoint via $\boldsymbol{\Omega}_{des} = \boldsymbol{\Omega} + \Delta t\, \mathbf{J}^{-1}\boldsymbol{\tau}^*$ (where $\Delta t = 4$ ms is the filter period), and thrust is normalized to $[-1, 0]$ as required by PX4's `thrust_body[2]` convention.

B. Physics Model

The custom physics environment propagates full 6-DOF rigid-body dynamics in the East-North-Up (ENU) frame, where $\mathbf{g}_{\text{ENU}} = [0, 0, -g]^T$ is the gravitational acceleration vector and $\Delta t = 4$ ms is the integration time step:

$$m\ddot{\mathbf{p}} = f\mathbf{R}\mathbf{e}_3 + m\mathbf{g}_{\text{ENU}}, \quad \mathbf{J}\dot{\boldsymbol{\Omega}} = \boldsymbol{\tau} - \boldsymbol{\Omega} \times \mathbf{J}\boldsymbol{\Omega}, \tag{18}$$

The inner angular rate loop is modeled as a first-order tracker with bandwidth $\omega_{BW} = 200$ rad/s:

$$\dot{\boldsymbol{\Omega}} = \omega_{BW}(\boldsymbol{\Omega}_{des} - \boldsymbol{\Omega}). \tag{19}$$

This gives an Euler eigenvalue $\lambda = 1 - \omega_{BW}\Delta t = 0.2$ per step, approximating the closed-loop inner-loop bandwidth of the x500 airframe (≈ 20 Hz crossover). The rotation matrix $\mathbf{R}$ is re-orthonormalized at every step via SVD to prevent

numerical drift. Attitude and odometry are published in the North-East-Down (NED) frame following PX4 convention.

Vehicle parameters for the x500 quadrotor are listed in Table IV.

TABLE IV

X500 SITL VEHICLE PARAMETERS AND CBF FILTER SETTINGS.

Parameter	Symbol	Value
Vehicle dynamics (x500 airframe)		
Mass	m	1.5 kg
Inertia (diag.)	$\mathbf{J}$	$[0.029,\ 0.029,\ 0.055]$ kg m^2
Min thrust	$f_{\min}$	0 N
Max thrust	$f_{\max}$	38 N ($\approx 2.5\,g$)
Max body torque	$\tau_{\max}$	2.5 N m
CBF filter		
FoV half-angle	$\Phi_{\max}$	25°
Boresight (body)	$\mathbf{v}_B$	$[1, 0, 0]^T$
ECBF gains	c_1, c_2	8, 16
Cost weights	$\mathbf{W}$	diag(1, 10, 10, 10)
Slack penalty	p_1	10^5
ADMM penalty	ρ	100
Filter rate		250 Hz
Lee geometric controller (mission node)		
Position gain	k_p	1.5 N m^{-1}
Velocity gain	k_v	3.0 N s m^{-1}
Attitude gain	k_R	4.0 N m rad^{-1}
Rate gain	k_Ω	0.5 N m s rad^{-1}

C. Mission Design and Yaw Policy

The pursuit scenario consists of four waypoints toward a target at $\mathbf{p}_T = [20,\ 6,\ 3]^T$ m (ENU), listed in Table V. The total armed flight time is 60 s. The controller gains $k_p = 1.5$, $k_v = 3.0$ yield a damping ratio $\zeta = k_v/(2\sqrt{k_p m}) = 1.0$, ensuring a critically-damped translational response.

TABLE V

WAYPOINT SEQUENCE (ENU, METERS).

WP	East	North	Up	WP	East	North	Up
0	0	0	2	2	8	5	3.5
1	6	0	2	3	14	6	3

The yaw reference is set to point the boresight $\mathbf{v}_B$ continuously toward the target $\mathbf{p}_T$ rather than toward the current waypoint. When the nose tracks the next waypoint instead, the boresight can deviate 30°–45° from the target azimuth during lateral segments, immediately violating the 25° FoV constraint. With yaw-toward-target, the main source of violations is pitch-axis tilt during acceleration and deceleration phases. The two scenarios probe complementary failure modes: yaw-toward-waypoint in the $SE(3)$ simulation and yaw-toward-target here.

D. CBF Invariant Results

Table VI and Fig. 3 compare the CBF-filtered and unfiltered runs. The ADMM-CBF node achieves zero FoV violations over all 14,500 armed steps ($h_{\min} = +0.040$), confirming that the safe set $\mathcal{S}$ is kept forward-invariant. The unfiltered baseline incurs 40 constraint violations (0.3% of steps), with $h_{\min} = -0.019$. The violations cluster around the WP0→WP1 and WP2→WP3 transitions, where backward pitch during deceleration momentarily swings the boresight above the target elevation.

TABLE VI

CUSTOM 6-DOF SITL: CBF VS. UNFILTERED BASELINE (60 S, $\Phi_{\max} = 25°$).

Method	$\min h_{FoV}$	Pitch (°)	|roll| (°)	Viol. (%)
PX4 Unfiltered	-0.019	28.1	19.2	0.3
PX4 + ADMM-CBF	$+0.040$	20.0	24.0	0.0

A notable feature of Table VI is the exchange of pitch and roll: the CBF-filtered vehicle shows higher maximum roll (24.0° vs. 19.2°) and lower maximum pitch (20.0° vs. 28.1°). This is consistent with the filter redirecting corrective action toward the roll axis, which does not violate the LOS cone constraint in this yaw-toward-target scenario, rather than allowing pitch to grow unchecked.

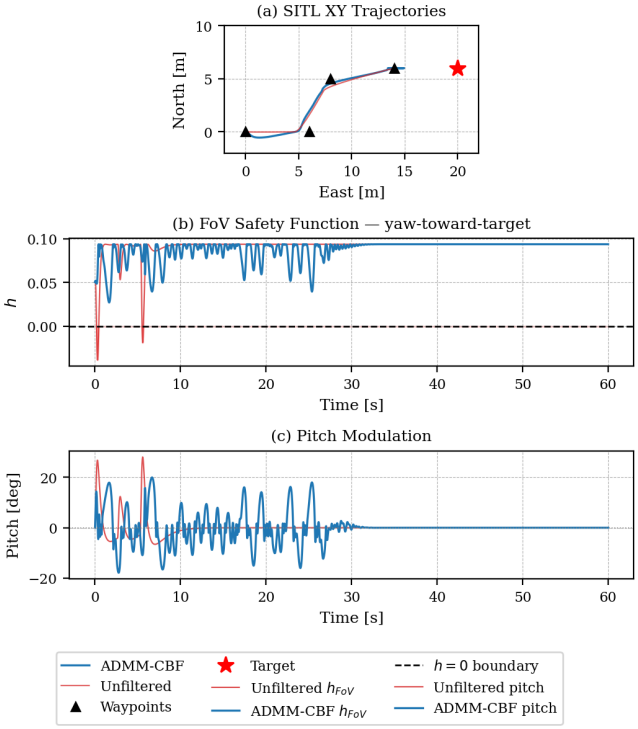


Fig. 3. Custom 6-DOF SITL results (60 s, x500 airframe, yaw-toward-target). (a) XY trajectories for both runs; waypoints are marked with triangles and the directed-energy target with a star. (b) FoV safety function h_{FoV}: the unfiltered baseline (red) dips below zero 40 times during deceleration phases at WP transitions ($h_{\min} = -0.019$); the CBF (blue) remains strictly positive ($h_{\min} = +0.040$). (c) Pitch evolution: the CBF limits backward pitch to 20.0°, whereas the unfiltered controller reaches 28.1°.

E. Computational Performance

Table VII reports wall-clock timing for the CBF node on an Intel i7-8700 (3.2 GHz, Python). The Cholesky factorization and triangular back-solves are $\mathcal{O}(1)$ per ADMM

iteration; the variability in total step time (P95 $= 832\,\mu$s, max $= 6{,}748\,\mu$s) reflects only the variable ADMM iteration count, not any structural change in the per-iteration cost. The worst-case total occurs at the transition from initialization to armed flight, when the warm-start cache is empty. No further budget overruns are observed in steady state. These timings are pure in-process measurements. The uXRCE-DDS bridge adds approximately 1–2 ms of transport latency to the flight controller but does not affect the CBF's computational guarantees.

TABLE VII

CBF NODE WALL-CLOCK TIMING (ARMED PHASE, INTEL I7-8700, PYTHON). P95 DENOTES THE 95TH PERCENTILE.

Operation	Mean (μs)	Std (μs)	P95 (μs)	Max (μs)
Lie derivative eval	191	64	207	1,769
ADMM solve	137	178	506	5,525
Total filter step	347	209	832	6,748
ADMM iterations	mean ≈ 4		P95 $= 26$	max $= 50$

A C or C++ reimplementation of the 5×5 Cholesky and triangular solve would reduce the mean total to well below $50\,\mu$s, consistent with TinyMPC microcontroller results [14].

VII. CONCLUSION

We developed a hardware-aware safety filter that enforces the electronic steering limits of a directed energy phased array payload on a quadrotor interceptor. The filter uses a relative-degree-two Control Barrier Function on the $SE(3)$ manifold. Its analytical Lie derivative chain produces a closed-form ECBF constraint, and a constant-time Cholesky factorization on a fixed 5×5 system enables a warm-started ADMM solver with microsecond mean execution.

The proposed filter achieves zero FoV violations in both nonlinear $SE(3)$ simulation and a custom closed-loop 6-DOF SITL environment. In contrast, conventional pitch clamping provides negligible improvement against coupled roll/yaw dynamics. An MPC-CBF extension ($N = 10$ look-ahead steps) exhibits a fundamental limitation of single-control multi-step safety filtering: open-loop rollout pessimism forces all $N + 1$ simultaneous ECBF constraints to demand recovery at once, causing actuator saturation. The ADMM-CBF avoids this with a single reactive constraint on a fixed factorization, achieving zero violations without mission degradation.

A $\Phi_{\max}$ sweep demonstrates the filter safely tightens the beam down to 15°, recovering up to 12.0 dB of antenna gain (a $16\times$ reduction in required transmit power) with mean solve time of 347 μs at 250 Hz.

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